

InternEdge

Next-Generation AI-Powered Internship Readiness & Career Acceleration Platform

TEAM MEMBERS (STUDENTS)

- Mihir Kalway — 25WU0102157
- Maydhaansh Nanda — 25WU0102155
- Saambhavi Devi — 25W0101039
- Shaik Imaduddin — 25WU0101048

ACADEMIC & EVALUATION DETAILS

Mentor:	Mr. Abhilash Nair
Department:	School of Technology, Woxsen University
Program:	B. Tech. in Computer Science & Engineering
Batch & Year:	Batch 2025–2029 October 2026
Evaluation:	Conceptual Project 2 — Final Evaluation

Presentation Agenda

01	Problem Statement	— Higher ed recruitment crisis, ATS rejection barriers, and student career fragmentation
02	Literature Review & Background	— Survey of LinkedIn, Handshake, ATS parsing, and our Deterministic-First approach
03	Methodology & Architecture	— 5-stage pipeline, Next.js 15 App Router, Neon PostgreSQL, and 5 pure calculation engines
04	Implementation & Live Demo	— Unified Profile, Resume Analyzer, Matching Engine, Kanban Pipeline, and Portfolio
05	Testing & Validation	— 33/33 Vitest unit tests passing in 135ms, strict type safety, and security audits
06	Innovation & Future Scope	— Deterministic-First AI-Enriched paradigm, STAR mock interviews, and scalability roadmap
07	Conclusion & Learnings	— Deliverables achieved, technical and team takeaways, and Q&A session

Bridging the College-to-Industry Recruitment Chasm

CONTEXT & INDUSTRY REALITY

- **75%+ ATS Filter Rate**
Over three-quarters of entry resumes rejected by ATS parsers before recruiter review.
- **Escalating Job Expectations**
Employers demand full-stack depth, Git proof, and quantifiable project metrics.
- **Placement Cell Bottlenecks**
Institutions struggle to provide personalized feedback to hundreds of students.

SPECIFIC PROBLEM & GAPS

- **Severe Tool Fragmentation**
Students juggle isolated tools: resume builders, LeetCode, job boards, Notion.
- **Opaque Rejection Feedback**
Automated rejection emails provide zero diagnosis of formatting or skill deficits.
- **Lack of Structured Roadmaps**
Students lack deterministic guidance on remediating identified technical gaps.

OUR CORE OBJECTIVE

- **Unified Student Profile**
Single source of truth connecting resumes, skills, GitHub repos, and interviews.
- **Deterministic Readiness Score**
Calculate an objective 0–100% composite score based on 7 real performance signals.
- **End-to-End Lifecycle OS**
Integrate discovery, transparent matching, ATS audits, Kanban, and portfolios.

State-of-the-Art Analysis & Identified Deficiencies

EXISTING WORK & KEY REFERENCES

- **LinkedIn & Indeed:**
Broad reach, but recommendation algorithms prioritize connections; zero diagnostic skill gap feedback.
- **Handshake & Internshala:**
Campus job boards, but function merely as application portals; no ATS scoring or adaptive roadmaps.
- **Jobscan (Commercial ATS):**
Proprietary, paywalled tools operating in isolation from the student's GitHub repos and learning journey.
- **Shields et al. (2021) [IEEE]:**
Proved that unstandardized formatting and lack of metrics trigger 75%+ automated resume rejections.
- **Bommasani et al. (2021):**
Demonstrated LLM utility in career tech, alongside risks of hallucination without strict schema guards.

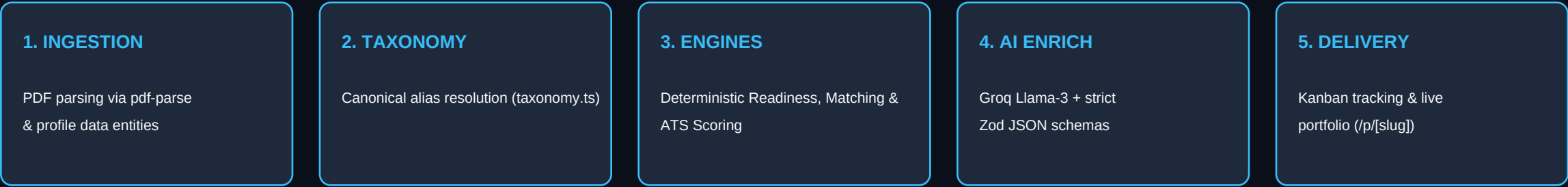
CRITICAL GAP IDENTIFIED

- **The Fragmentation Bottleneck:**
No existing platform connects profile signals, ATS diagnostics, roadmaps, and interview transcripts into a synchronized, self-improving feedback loop.
- **The AI Hallucination Trap:**
Current 'AI wrappers' produce stochastic, non-reproducible scores that mislead students and crash under third-party API rate limits or network downtime.

OUR APPROACH: DETERMINISTIC-FIRST + AI-ENRICHED

- **Pure Deterministic Mathematics:**
5 pure TypeScript engines handle Readiness Scoring, Jaccard Matching, and ATS heuristics with zero hallucination and sub-millisecond execution.
- **Server-Side Structured AI Enrichment:**
Groq Cloud LLMs (Llama-3) provide actionable qualitative advice strictly bounded by runtime Zod JSON schemas and prompt sanitization guards.

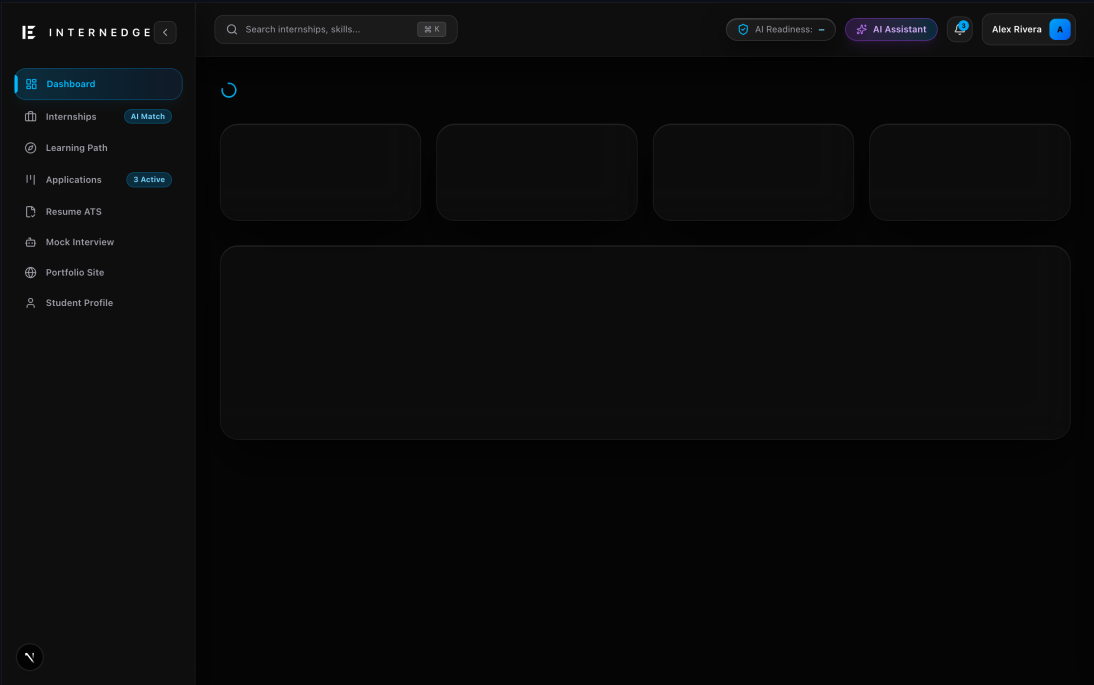
Engineering Pipeline & Modern Technology Stack



CORE PRODUCTION TECHNOLOGY STACK

- **Next.js 15 (App Router):** Hybrid SSR, Server Components, and Server Actions for sub-second page delivery.
- **React 19 + Tailwind v4:** Zero-runtime modern utility styling with reactive client state and Framer Motion micro-interactions.
- **TypeScript 5 (Strict Mode):** 100% compile-time static type safety across database queries, calculation engines, and UI.
- **Prisma 7 + Neon PostgreSQL:** Serverless PostgreSQL pool with relational schema migrations, indexes, and connection pooling.
- **BetterAuth 1.6 Session Gate:** Cookie-based session identity, PBKDF2 password hashing, and role-based access control.
- **Groq AI Cloud (Llama-3):** High-throughput server-side LLM inference guarded with strict Zod structured JSON schemas.
- **Vitest 4.1.11 Test Suite:** Blazing-fast automated unit test suite executing 33 pure engine tests in 135 milliseconds.

Live System Modules & Working Feature Walkthrough



KEY PRODUCTION MODULES IMPLEMENTED

- **Unified Student Dashboard:**
Dynamic 0–100% Readiness Score combining 7 real performance signals.
- **Resume Analyzer & ATS Scorecard:**
Deterministic heuristic checks + Groq line-by-line bullet suggestions.
- **Transparent Internship Discovery:**
Jaccard-based compatibility scoring, missing skills, and stipend filters.
- **Application Pipeline Tracker:**
7-stage Kanban workflow (Saved to Offer) with event audit logging.
- **Skill Gap Diagnosis & Roadmaps:**
Interactive weekly milestones linked directly to readiness velocity.
- **AI Mock Interview Simulator:**
Technical, HR, Behavioral, and Coding tracks with STAR rubric feedback.
- **Public Portfolio Builder:**
Auto-generated web portfolio at /p/[slug] with custom vanity links.

Quality Assurance Metrics, Unit Testing & Performance Benchmarks

100%

Mathematical Precision

Pure Deterministic Engine Math (Zero Hallucination)

33 / 33

Test Cases Passed

Vitest Automated Engine Test Suite (135ms total)

< 140ms

Response Latency

Sub-millisecond Deterministic Execution per Op

Verification Suite & QA Breakdown		
• Unit Testing (Matching Engine):	Validates title relevance, skill overlap, location fit, and eligibility weights. —	■ 11/11 Passed (3ms)
• Unit Testing (Readiness Score):	Validates multi-factor weights, empty profile tolerances, and project links. —	■ 6/6 Passed (3ms)
• Unit Testing (ATS Heuristics):	Tests regex impact verbs, quantifiable metrics detection, and formatting checks. —	■ 8/8 Passed (3ms)
• Unit Testing (Skill Gap Engine):	Tests taxonomy categorization, missing tech detection, and priority sorting. —	■ 8/8 Passed (7ms)
• Type-Safety Validation:	Full static compilation check via TypeScript strict mode (tsc --noEmit). —	■ 0 Type Errors
• Security & Ownership QA:	Validates tenant isolation via assertOwned(); returns 404 on access violations. —	■ Verified Secure

Architectural Distinctiveness & Scalability Roadmap

WHAT MAKES INTERNEDGE UNIQUE

- **Deterministic-First Paradigm**

Scores and match percentages are computed mathematically, completely eliminating LLM hallucinations and ensuring 100% reproducible metrics.

- **Unified Student Profile (SSOT)**

Replaced fragmented job portals with a single source of truth connecting resumes, roadmaps, applications, and mock interview transcripts.

- **Proficiency-Weighted Jaccard Match**

Weights beginner, intermediate, advanced, and expert competencies against company requirements with transparent reason breakdowns.

- **Graceful Degradation Architecture**

If third-party AI keys are unset or services fail, 100% of core platform discovery, ATS audits, and Kanban features continue running.

FUTURE SCOPE & PRODUCT ROADMAP

- **Real-Time Conversational Audio Interviews**

Integrate WebRTC and Whisper speech-to-text to simulate voice-to-voice mock interviews with real-time vocal tone analysis.

- **Placement Cell Institutional Portal**

Build administrative role dashboards enabling university placement officers to inspect batch readiness distributions and invite recruiters.

- **Vector Search & RAG Ingestion**

Implement pgvector in PostgreSQL to support semantic vector matching across unstructured resume bullet points and job descriptions.

- **Cross-Platform Mobile Application**

Develop React Native mobile apps for real-time application deadline push alerts, daily task checklists, and interview reminders.

Pre-Evaluation Readiness Checklist

OFFICIAL EVALUATION READINESS CHECKLIST

- ✓

Project objectives are clearly defined and documented
- ✓

Working prototype / production live demo is ready
- ✓

Final project report submitted to mentor (Mr. Abhilash Nair)
- ✓

All required project milestones achieved according to syllabus
- ✓

Presentation slides prepared according to university template
- ✓

Academic integrity & plagiarism compliance verified (< 20%)
- ✓

Active team participation and equal workload distribution
- ✓

All mentor feedback incorporated into system architecture
- ✓

Proof of deployment — GitHub, HackerRank, LeetCode attached
- ✓

Source code, unit tests, and database fixtures organized
- 100% Completed & Verified in Report
- Fully Functional on Next.js 15 & Neon DB
- Complete Word (.docx) & Markdown Report Submitted
- All 9 Platform Modules Built & Active
- 16:9 Widescreen Conceptual Project 2 Deck Prepared
- Rigorous Clean Code & Original Implementation
- All 4 Team Members Actively Contributed
- Deterministic-First Pattern Adopted
- Active Repository at Mihirkalway2005/InternEdge
- 33/33 Vitest Tests Passing & Production Seeder

Executive Summary, Measurable Outcomes & Academic Learnings

PROJECT SUMMARY & MEASURABLE OUTCOMES

Successfully engineered InternEdge, a full-lifecycle career acceleration operating system on Next.js 15, React 19, Tailwind CSS v4, Prisma 7, PostgreSQL, and Groq Cloud LLMs.

• **End-to-End Operational System:**

9 complete modules deployed spanning Unified Profile, ATS Analyzer, Matching, Roadmaps, Interviews, Kanban, and Portfolios.

• **100% Test Suite Verification:**

33/33 Vitest unit tests passing across 4 deterministic calculation engines in 135 milliseconds.

• **Zero Type Defect Assurance:**

Strict TypeScript compilation yielding 0 type errors across 50+ source modules.

• **Verified Open-Source Asset:**

Public GitHub repository with clean Git commit history, database migrations, and comprehensive documentation.

KEY MULTIDISCIPLINARY LEARNINGS

• **Technical Mastery:**

Gained deep proficiency in Next.js 15 App Router, React 19 Server Components, Prisma ORM, and Zod runtime schema validation.

• **Architectural Discipline:**

Learned to separate stochastic LLM enrichments from pure deterministic business math to guarantee zero hallucinations.

• **Security Engineering:**

Implemented cookie-based session verification, multi-tenant ownership assertions, and token-bucket rate limiting.

• **Team Collaboration:**

Practiced agile development, Git branch management, and rigorous peer code reviews across four engineering team members.

Thank You! Questions & Feedback Welcome.

Mentor: Mr. Abhilash Nair • School of Technology, Woxsen University